

RC Design Project 01

We can delete this since it no need

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1 Introduction

This is Our Team Intro.

2 Detailed Design Calculations

2.1 Load

For imposed loads on floor according to Table 6.1 – 6.4 in EN 1991-1-1:2002 [3]. Is summary in Table 2.1.

Table 2.1: Imposed Loads On Floor Base On EN 1991-1-1:2002

Category	Sub-category / Use	q_k [kN/m ²]	Q_k [kN]
Category A	Areas for domestic and residential activities		
	- Floors	1,5 to 2,0	2,0 to 3,0
	- Stairs	2,0 to 4,0	2,0 to 4,0
	- Balconies	2,5 to 4,0	2,0 to 3,0
Category B	Office areas	2,0 to 3,0	1,5 to 4,5
Category C	Areas where people may congregate		
C1	Areas with tables (e.g. schools, cafes, restaurants, reading rooms)	2,0 to 3,0	3,0 to 4,0
C2	Areas with fixed seats (e.g. theatres, cinemas, churches)	3,0 to 4,0	2,5 to 7,0 (4,0)
C3	Areas without obstacles for moving people (e.g. museums, entrance halls)	3,0 to 5,0	4,0 to 7,0
C4	Areas with possible physical activities (e.g. dance halls, gyms)	3,5 to 5,0	3,5 to 7,0
C5	Areas susceptible to large crowds (e.g. concert halls, sports halls with stands)	5,0 to 7,5	3,5 to 4,5
Category D	Shopping areas		
D1	Areas in general retail shops	4,0 to 5,0	3,5 to 7,0 (4,0)

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Table 2.1: Imposed Loads On Floor Base On EN 1991-1-1:2002

Category	Sub-category / Use	q_k [kN/m ²]	Q_k [kN]
D2	Areas in department stores	4,0 to 5,0	3,5 to 7,0
Category E	Storage and industrial activities (general storage, accessible areas)	7,5	7,0

And this is the imposed loads on second floor that we assumed. which shown in Table 2.2.

Table 2.2: Imposed Loads On Second Floor

No.	Room / Area	Category (EN 1991-1-1)	q_k (kN/m ²)
1	Receptionist/ Reception	B	3
2	Workstation / Open office	B	3
3	Meeting Room	C1	3
4	President Office	B	3
5	Vice President Offices	B	3
6	Accounting	B	3
7	General Sale & Administration	B	3
8	Copy / Printer Area	B	3
9	Storages	E1	5
10	Lift Lobby / Lift Area	C3	4
11	Stair (Main Stair)	A	4
12	Restroom (Men)	B	3
13	Restroom (Women)	B	3
14	—	—	—
15	—	—	—
16	—	—	—
17	Entrance	C3	4
18	Emergency Stair	A	4

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Table 2.2: Imposed Loads On Second Floor

No.	Room / Area	Category (EN 1991-1-1)	q_k (kN/m ²)
19	Server& Technical Room	E1 (conservative)	5
20	Kitchen / Pantry	C1	3
21	Library	C3	4
22	Share Space	C3	4

References

- [1] BSI, *BS EN 1992-1-1:2023 - Eurocode 2: Design of concrete structures - Part 1-1: General rules and rules for buildings, bridges and civil engineering structures*, BSI Standards Publication, 2023.
- [2] BSI, *BS EN 1991-1-1:2025 - Eurocode 1: Actions on structures - Part 1-1: Specific weight of materials, self-weight of construction works and imposed loads for buildings*, BSI Standards Publication, 2025.
- [3] British Standards Institution, *BS EN 1991-1-1:2002 - Eurocode 1: Actions on structures - Part 1-1: General actions - Densities, self-weight, imposed loads for buildings*, Incorporating corrigenda December 2004 and March 2009, BSI, 2002.
- [4] CEN, *EN 1991-1-4:2005+A1 - Eurocode 1: Actions on structures - Part 1-4: General actions - Wind actions*, European Standard, Incorporating corrigendum January 2010, April 2010.
- [5] CEN, *Eurocode 1: Actions on structures - Part 1-4: General actions - Wind actions*, EN 1991-1-4:2005+A1, English version, April 2010.

Note that : We may use the referennces provide by professor (check in telegram group) if it still not enough we can add more maybe.

Appendices

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